

NewsBot Intelligence System 2.0

Executive Summary

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github.com/Tikskalani/ITAI2373-Portfolio

The opportunity

Organizations are buried in news. Communications teams, investor-relations groups, and market analysts need to know, within minutes, what is being said about their company and their sector, whether the tone is turning, and who is being named. Reading feeds by hand does not scale, and generic keyword alerts miss context: they cannot tell a positive story from a negative one, cannot group related coverage, and cannot answer a plain-language question.

The solution

NewsBot Intelligence System 2.0 is a modular natural language processing platform that turns raw news text into structured, decision-ready intelligence. For any article it returns the category with a confidence score, the sentiment and dominant emotion, the named people, organizations, and places, and a short summary. Across a collection it discovers the underlying topics and finds related coverage by meaning rather than exact keywords. A conversational layer lets a non-technical user simply ask questions such as "what is the sentiment?" or "who is mentioned?", and a web dashboard puts all of this in a browser.

Value proposition

- **Faster awareness.** Coverage is categorized, scored, and summarized automatically, so a team sees what matters without reading every article.
- **Tone monitoring.** Sentiment and emotion scoring flags when coverage starts to turn negative, the moment that matters most for a communications team.
- **Searchable intelligence.** Entity extraction turns a pile of articles into a searchable index of the people, companies, and places in the news.
- **Trustworthy by design.** When input is out of scope, the system reports 'uncertain' instead of guessing, so users are not misled by a confident wrong answer.
- **Accessible.** The conversational interface and web dashboard mean non-technical staff can use it without writing code.

Illustrative ROI

The figures below are an illustration of the time-savings case, not audited results. They assume an analyst who currently triages news by hand.

Task	Manual	With NewsBot	Saving
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Triage 200 articles/day	~3.5 hours	~20 minutes	~3 hours/day
Flag negative coverage	reactive, hours later	near real time	hours earlier
Build an entity index	manual, rarely done	automatic	new capability
Annual analyst time (1 FTE)	reference	~60% of triage time freed	reallocated to analysis

The core value is reallocating skilled analyst time away from manual triage and toward judgment and response.

Use cases

- **Corporate communications:** monitor brand mentions and alert on negative sentiment spikes.
- **Investor relations and finance:** track how a company or sector is being covered and surface the named entities driving a story.
- **Media and research teams:** cluster a day's coverage into topics and pull the most relevant articles on demand.
- **Multilingual monitoring:** detect and translate non-English coverage into the same analysis pipeline.

Competitive landscape

Approach	Categorize	Sentiment	Entities	Topics	Ask in plain English	Honest uncertainty
Keyword alerts	No	No	No	No	No	n/a
Generic dashboards	Partial	Basic	Partial	Rare	No	No
Enterprise suites	Yes	Yes	Yes	Some	Some	Rare
NewsBot 2.0	Yes	Yes + emotion	Yes	LDA + NMF	Yes	Yes

NewsBot 2.0 is a course capstone, not a commercial product. It is positioned against these categories to show where its design choices, especially explainable uncertainty and a conversational interface, add value.

Honest limitations

The classifier is trained on the BBC News corpus, so it performs best on general news in that style and is weaker on messy present-day text. Lexicon sentiment misreads financial framing and would be replaced with a finance-tuned model before any real deployment. These limits are stated plainly because trustworthy tooling depends on knowing where it is wrong.

Bottom line

NewsBot 2.0 demonstrates a production-style NLP system that saves analyst time, monitors tone, and answers questions in plain language, while being honest about its limits. It is portfolio-quality evidence of end-to-end applied NLP skill, from data and modeling through a tested, documented, deployable application.